

Creating new VCN in OCI

Step 1: Log in to Oracle Cloud Console

1. Visit <https://cloud.oracle.com>
2. Log in with your credentials.

Step 2: Navigate to the VCN Wizard

1. From the **main menu** (☰), go to **Networking** → **Virtual Cloud Networks**.
2. Click on **Start VCN Wizard**.

Step 3: Choose a VCN Creation Option

You'll see three options:

- **VCN with Internet Connectivity** (Recommended for most general uses)
- **VCN with Public and Private Subnets**
- **VCN with Only Private Subnets**

For this guide, select **VCN with Internet Connectivity** and click **Start**

Step 4: Configure VCN and Subnet Settings

1. **VCN Name:** Enter a name for the VCN (e.g., **MyProjectVCN**).
2. **Compartment:** Choose the appropriate compartment.
3. **CIDR Block:** Default is usually **10.0.0.0/16** – adjust if needed.
4. **DNS Resolution:** **Enable**
5. **Subnet Settings:**
 - **Public and Private Subnet Name:** (e.g., **MyPublicSubnet**)
 - **CIDR Block for Public and Private:** (e.g., **10.0.0.0/24**)
6. Review and create

Steps to Create and Setup a Private IP Instance

Step 1: Login to Oracle Cloud Console

- Go to [Oracle Cloud](#)
 - Login with your credentials
 - Navigate to **Compute > Instances**
-

Step 2: Create the Private IP Instance

- Click **Create Instance**
 - Follow the same steps used to create a public IP instance with the following key differences:
 - **VCN:** Select the **newly created VCN**
 - **Subnet:** Select a **Private Subnet** (ensure it has no public IP assignment)
 - Configure other options as required
-

Step 3: Configure SSH Access

- Upload your **Public Key** or generate a new key pair
 - Download and securely save your **Private Key**
 - Click **Create** to launch the instance
-

Step 4: Enable Console Access

- Go to the instance **Details Page**
- Under **Resources**, click on **Console Connections**
- Create a new serial console connection if not already available
- Use the downloaded private key to connect to the instance via serial console
- Login as **root**

Step 5: Configure Network Access

1. **Change SSH Port (if required) to match your VPN server's configuration.**

2. **Flush existing iptables rules:**

```
iptables -F
```

3. **Set iptables to allow VPN traffic:**

```
iptables -A INPUT -p tcp --dport <VPN_PORT> -j ACCEPT
```

4.

```
iptables -A FORWARD -j ACCEPT
```

5.

```
iptables -A INPUT -j ACCEPT
```

(Replace <VPN_PORT> with your actual VPN port, e.g., 1194)

6. **Allow VPN port in UFW (if UFW is enabled):**

```
ufw allow <VPN_PORT>/tcp
```

Step 6: Test Connectivity

- Ensure your VPN server is up and connected
 - Try connecting to the private instance from your local machine via VPN
 - You should be able to SSH using the private IP of the instance
-

Step-by-Step Setup for Remote VCN Peering (Upgraded DRG) – With two different tenancy

1. Prerequisites Check

- Ensure **both regions are subscribed** in your tenancy.
- Confirm **VCN CIDRs don't overlap** (e.g., VCN-A: 10.0.0.0/16, VCN-B: 192.168.0.0/16).
- Create or confirm **Virtual Cloud Networks (VCNs)** exist in each region.

2. Create DRGs and Attach to VCNs

In Region 1:

- Go to **VCN** → **Under Resource** → **DRG** → **Create DRG** (name it **DRG-A**).
- Attach **DRG-A** to **VCN-A**.

In Region 2:

- Go to **VCN** → **Under Resource** → **DRG** → **Create DRG** (name it **DRG-B**).
- Attach **DRG-B** to **VCN-B**.

Resources

Subnets (2)

CIDR Blocks/Prefixes (1)

Route Tables (2)

Internet Gateways (1)

Dynamic Routing Gateways Attachments (1)

Network Security Groups (0)

Security Lists (2)

Dynamic Routing Gateway Attachments

VCNs are connected to a DRG by an attachment with a VCN type. You can configure all VCNs attached to the DRG to use the same route table.

The VCN has reached the maximum number of DRG attachments. You must remove an existing attachment to create a new attachment.

Create DRG Attachment

Attachment Name	Lifecycle State	Dynamic Routing Gateway	VCN Route Table	Cross-Tenancy	Created
KRG-DC-DRG	● ATTACHED	DRG-A	—	No	Fri, May 9, 2025, 04:48:56 UTC

Showing 1 item < 1 of 1 >

3. Create Remote Peering Connections (RPCs)

In Region 1 (Requestor):

- Go to **DRG-A** → **Remote Peering Connections** → **Create RPC**
 - Name it: **RPC-A**
 - Choose: **Requestor**

In Region 2 (Acceptor):

- Go to **DRG-B** → **Remote Peering Connections** → **Create RPC**
 - Name it: **RPC-B**
 - Choose: **Acceptor**

Resources

VCN attachments (1)

Virtual circuit attachments (0)

IPSec tunnel attachments (0)

Remote peering connection attachments (1)

Loopback attachments (0)

Cross-tenancy attachments (0)

Remote peering connection attachments in ptpl (root) *Compartment*

Remote peering connection (RPC) attachments are automatically created when an RPC is created. You can't directly create additional attachments for an RPC.

Create remote peering connection

Attachment name	Lifecycle state	DRG route table	Remote peering connection	Peering status	Created
DRG Attachment for RPC: RPC-A	Attached	Autogenerated DRG Route Table for RPC, VCN, and IPSec attachments	RPC-A	Peered	Fri, May 9, 2025, 04:57:04 UTC

Showing 1 item < 1 of 1 >

Create remote peering connection Help

Name

Create in compartment

ptpl (root)

This creates an attachment to the selected DRG. The attachment uses a route table based on the type of resource using the attachment.

Show Advanced options

Create remote peering connection

Cancel

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What's this? | Redwood preview

4. After creating RPC, Go to RPC A in the fourth column of the shown above screenshot

5. Click on RPC A , There you will see the Establish Connection box Click on that

RPC

AVAILABLE

Establish connection

Edit

Terminate

Remote peering connection Information

Compartment: ptpl (root)

DRG OCID: ...e552hwkckq [Show](#) [Copy](#)

Peer status: Peered

Peer region: me-dubai-1

Peer connection OCID: ...3g5j56c4ra [Show](#) [Copy](#)

OCID: ...vbbjwgbg [Show](#) [Copy](#)

Created: Fri, May 9, 2025, 04:56:59 UTC

Cross-tenancy: No

Peer tenancy OCID: ...62qw7xZiq [Show](#) [Copy](#)

Resources

Remote peering connection attachments (1)

Remote peering connection attachments

Attachment name	Lifecycle state	Dynamic routing gateways	DRG route table	Created
DRG Attachment for RPC: RPC-A	Attached	DRG-A	Autogenerated DRG Route Table for RPC, VCN, and IPSec attachments	Fri, May 9, 2025, 04:57:04 UTC

Showing 1 item < 1 of 1 >

4. Establish the Peering Connection

In Region 1 (Requestor):

- Go to **RPC-A** → **Establish Connection**
- Enter:
 - **RPC-B's** → **Region**
 - **RPC-B's OCID**
- Click **Establish**.

Connection will show as "**Peered**" once successful.

In Region 2 (Acceptor):

- Go to **RPC-A** → **Establish Connection**
- Enter:
 - **RPC-A's** → **Region**
 - **RPC-A's OCID**
- Click **Establish**.

Connection will show as "**Peered**" once successful.

OCID of RPC shown in below image → **LEFT block of DRG OCID and RIGHT block of RPC OCID**

RPC-A

Establish connection Edit Terminate

Remote peering connection Information

Compartment: ptpl (root)

DRG OCID: ...e52hwkckq [Show](#) [Copy](#)

Peer status: ● Peered

OCID: ocid1_remotepeeringconnection.oc1.ap-hyderabad-1.aaaaaaaypr67yslondsghtofjesengb3tbvK5vvxzvqozlulvbbjewgba [Copy](#)

Created: Fri, May 9, 2025, 04:56:59 UTC

Cross-tenancy: No

5: Update Security Lists on Both VCNs

To allow traffic between both VCNs, you must **edit the security lists** on each end.

A. Modify Security List on VCN-A (Hyderabad)

1. Go to Instance → Under Primary VNIC → Subnet → Private-subnet-<VNCname>

Primary VNIC

Public IPv4 address:

Private IPv4 address: 10.0.1.253

Route table: [route table for private subnet-Testing](#)

Network security groups: None [Edit](#) ⓘ

Subnet: [private subnet-Testing](#)

Private DNS record: Enable

Hostname: instance-20250303-1244

Internal FQDN: instance-20250303-1244... [Show](#) [Copy](#)

2. Under Security List edit the ingress and egress rule

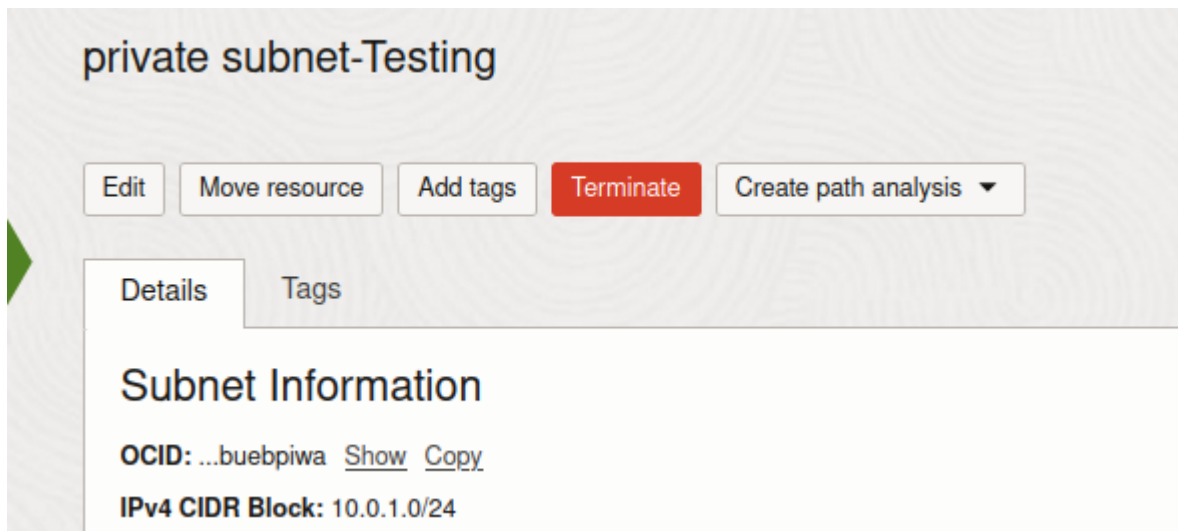
Security Lists

Add Security List

Name	State
security list for private subnet-Testing	● Available

3. Click **Add Ingress Rule**:

- **Source Type**: CIDR
- **Source CIDR**: (CIDR of VCN-B, e.g., **10.0.0.0/16**)



- **IP Protocol:** All Protocols (or specify TCP/UDP if needed)
- Click **Add Ingress Rule**

4. Click **Add Egress Rule:**

- **Destination Type:** CIDR
- **Destination CIDR:** (CIDR of VCN-B, e.g., **10.0.0.0/16**)
- **IP Protocol:** All Protocols
- Click **Add Egress Rule**

B. Modify Security List on VCN-B (Dubai)

1. **Switch back to me-dubai-1**
2. Go to **Virtual Cloud Networks**, open your VCN-B.
3. Scroll to **Security Lists** and click the one attached to your subnet.
4. Add the **Ingress Rule:**
 - **Source Type:** CIDR
 - **Source CIDR:** (CIDR of VCN-A, e.g., **192.168.0.0/16**)
 - **IP Protocol:** All Protocols
5. Add the **Egress Rule:**
 - **Destination Type:** CIDR

- **Destination CIDR:** (CIDR of VCN-A, e.g., **192.168.0.0/16**)
 - **IP Protocol:** All Protocols
-

8. Test the Peering

- Launch instances in both VCNs.
 - Ping or SSH from one to the other using private IPs.
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